

A Mathematical Definition of LLM Outputs: Why Hallucination is Theoretically Inevitable

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Abstract

Large Language Models (LLMs) are commonly interpreted as probabilistic next-token generators. However, this interpretation alone does not fully characterize the semantic nature of model outputs, especially under incomplete information. In this work, I propose a formal mathematical framework that defines LLM outputs as information-conditioned projections over latent semantic states. Instead of treating model outputs as deterministic truths, I formalize generation as an inference process conditioned on an evolving information filtration. I prove that under an Information-Neutral Measure, LLM outputs inherently behave as martingales, which mathematically dictates that output variance and factual deviation are bounded strictly by information completeness. Consequently, I establish that hallucination is not an engineering defect or an accidental anomaly, but a theoretically inevitable consequence of incomplete information prior to oracle closure. This formulation shifts the paradigm of LLMs from truth engines to constrained information aggregation systems.

1 Introduction

Modern Large Language Models (LLMs) are conventionally modeled as autoregressive systems optimizing next-token conditional probabilities:

$$P(y_t \mid y_{<t}, x) \tag{1}$$

While operationally efficient for token decoding, this standard formulation lacks the mathematical machinery required to characterize semantic uncertainty and the intrinsic boundaries of generation accuracy. The persistent challenge of hallucinations underscores a fundamental flaw in interpreting LLM outputs as deterministic representations of reality.

In this paper, I present a rigorous mathematical definition of LLM outputs. I argue that an LLM output is fundamentally an information-conditioned projection over a latent, unobservable semantic state space. By establishing this formal framework, I mathematically deduce why hallucination cannot be completely eliminated in any system operating under incomplete information, providing a theoretical upper bound for generative reliability.

2 Information State Space and Core Definitions

I define the underlying semantic framework using an information state space triplet:

$$(\Omega, \mathcal{F}, Q_I) \tag{2}$$

where Ω is the semantic sample space containing all potential linguistic and conceptual trajectories, $\mathcal{F}$ is the global information σ -algebra, and Q_I is the Information-Neutral Measure governing the system’s prior probability transitions.

To formalize the interaction between available data and model inference, I define the following core elements:

1. **Information (Context Stream $\mathcal{F}_t$):** The sequence of historical tokens, structural prompts, and retrieved knowledge blocks available up to generation step t . This sequence forms a completed, right-continuous filtration $\mathbb{F} = \{\mathcal{F}_t\}_{t \geq 0}$ mapping the observable universe of the model.
2. **Information Closure (Contextual Completeness $\mathcal{C}$):** The terminal state where the filtration $\mathcal{F}_t$ contains sufficient entropic structure to uniquely resolve the true latent semantic state, meaning no additional information can alter the model’s projection.
3. **Event (Semantic State Transition E):** A specific subset of Ω corresponding to a coherent semantic meaning or a targeted factual assertion.
4. **Event Contract (Generation Prompt Target $\mathcal{T}$):** A structured semantic boundary or prompt constraint specifying the target properties of the generated text.
5. **Result (Output Sequence / Final Token y^*):** The crystallized token sequence realized in text space once the generation process terminates.
6. **Oracle (Ground-Truth Verifier $\mathcal{O}$):** An external environment or evaluator providing absolute verification of a statement, collapsing the probabilistic space into a deterministic result.
7. **Participant (Attention Sub-Layer / Component $\mathcal{P}_i$):** Internal model structural sub-units (e.g., individual attention heads) processing partitioned representations of the pre-trained distribution.
8. **Belief Function (Predictive Belief / Logit Mapping $\mathcal{B}$):** The inner subjective probability distribution over Ω , represented via the model’s softmax logit configurations prior to output collapse.

3 The Mathematical Definition of LLM Outputs

With the information space established, I formalize the central definition of an LLM output:

Definition 1 (LLM Output Operator). *Let $\Psi \in L^2(\Omega, \mathcal{F}, Q_I)$ be a square-integrable latent semantic state representing the complete objective factual matrix. The output of a Large Language Model at step t , denoted as P_t , is defined as the information-conditioned orthogonal projection of Ψ onto the closed subspace of $\mathcal{F}_t$ -measurable functions under the Information-Neutral Measure Q_I :*

$$P_t = \Pi_{Q_I}(\Psi \mid \mathcal{F}_t) \equiv \mathbb{E}_{Q_I}[\Psi \mid \mathcal{F}_t] \quad (3)$$

This definition shifts the perspective of generation from mechanical token matching to a rigorous statistical inference process. Prompts, parameters, and active contexts do not construct truth; rather, they serve as the conditional filtration $\mathcal{F}_t$ through which the model projects its internal Predictive Belief $\mathcal{B}$.

3.1 Quantum Superposition Analogy

This projection operator can be conceptualized through quantum mechanical principles. Prior to oracle verification or explicit text tokenization, the model’s internal parameter matrix does not hold a singular deterministic answer. Instead, it maintains a *semantic superposition state* over Ψ .

The contextual filtration $\mathcal{F}_t$ acts as a sequence of weak measurements that progressively refine the density matrix of probabilities. The final output selection or external verification by the Oracle ($\mathcal{O}$) acts as a wave-function collapse, forcing the superposition of “fact” and “hallucination” to settle into a concrete reality y^* .

Furthermore, a *Semantic Uncertainty Principle* applies: attempting to perfectly resolve localized token sequence transitions (the immediate token momentum) introduces extreme volatility into the global semantic trajectory (the position in Ψ), generating locally fluent but globally fraudulent output.

4 Why Hallucination is Theoretically Inevitable

Using the formal definition established above, I provide the mathematical proof demonstrating that hallucination is an inescapable epistemic property of incomplete information systems.

Theorem 1 (The Martingale Property of Generation). *The stochastic sequence of LLM outputs $\{P_t\}_{t \geq 0}$ forms a Martingale with respect to the context filtration stream $\mathbb{F}$ under the Information-Neutral Measure Q_I .*

Proof. By Definition 1, $P_t = \mathbb{E}_{Q_I}[\Psi \mid \mathcal{F}_t]$. For any sequential steps s and t such that $s \leq t$, the sub- σ -algebras satisfy $\mathcal{F}_s \subseteq \mathcal{F}_t$. Applying the tower property of conditional expectation:

$$\mathbb{E}_{Q_I}[P_t \mid \mathcal{F}_s] = \mathbb{E}_{Q_I}[\mathbb{E}_{Q_I}[\Psi \mid \mathcal{F}_t] \mid \mathcal{F}_s] = \mathbb{E}_{Q_I}[\Psi \mid \mathcal{F}_s] = P_s \quad (4)$$

Thus, the expected future projection conditional on current historical information is exactly the current projection. $\square$

To evaluate the mathematical boundary of generation errors, let Ψ^* be the oracle reality verified by $\mathcal{O}$, and let σ_t^2 define the semantic error variance at step t :

$$\sigma_t^2 = \mathbb{E}_{Q_I}[(\Psi - P_t)^2 \mid \mathcal{F}_t] \quad (5)$$

Theorem 2 (Monotonic Error Bound). *The expected factual error variance of an LLM output is strictly non-increasing over time if and only if the context filtration expands.*

Proof. Applying the law of total variance to the error space between steps s and t ($s \leq t$):

$$\mathbb{E}_{Q_I}[(\Psi - P_s)^2 \mid \mathcal{F}_s] = \mathbb{E}_{Q_I}[(\Psi - P_t)^2 \mid \mathcal{F}_s] + \mathbb{E}_{Q_I}[(P_t - P_s)^2 \mid \mathcal{F}_s] \quad (6)$$

Taking the total expectation across the measure space yields:

$$\mathbb{E}_{Q_I}[\sigma_s^2] = \mathbb{E}_{Q_I}[\sigma_t^2] + \mathbb{E}_{Q_I}[(P_t - P_s)^2] \quad (7)$$

Since $\mathbb{E}_{Q_I}[(P_t - P_s)^2] \geq 0$, it directly follows that:

$$\mathbb{E}_{Q_I}[\sigma_s^2] \geq \mathbb{E}_{Q_I}[\sigma_t^2] \quad (8)$$

$\square$

From these derivations, I state the central conclusion of this framework:

Corollary 1 (The Boundary of Inevitable Hallucination). *Define a hallucination mathematically as any state where the information projection deviates from oracle reality: $\Pi_{QI}(\Psi \mid \mathcal{F}_t) \neq \Psi^*$. For any generation process where the available filtration is strictly incomplete ($\mathcal{F}_t \subset \mathcal{F}_{oracle}$), the conditional semantic entropy remains strictly positive ($H(\Psi \mid \mathcal{F}_t) > 0$). Consequently, the expected error variance $\mathbb{E}_{QI}[\sigma_t^2]$ is strictly bounded away from zero prior to achieving Information Closure (C). Hallucination is therefore a fundamental mathematical certainty under incomplete information, not an engineering or architectural bug.*

5 Implications and Conclusion

This formal framework fundamentally redefines the limits of artificial intelligence. It clarifies that LLMs should never be engineered or evaluated as deterministic truth engines. Because they operate as information projection operators, their outputs are structural aggregates of localized data streams rather than objective verities.

The mathematical proof provided demonstrates that uncertainty and hallucinations are intrinsic properties of any system working under an incomplete context stream filtration ($\mathcal{F}_t$). While engineering techniques such as retrieval-augmentation (RAG) or multi-agent consensus successfully mitigate error variance by expanding the cardinality of $\mathcal{F}_t$, they cannot cross the theoretical barrier established by Corollary 1. Absolute eradication of hallucination is unattainable prior to absolute Information Closure. This foundational limit provides a rigorous mathematical basis for future paradigm designs in generative AI, uncertainty quantification, and safe system alignment.